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ARTICLE

Using fish hard-part microchemistry and genetics to quantify population impacts of low-use lock-and-dam structures on the Alabama River

Garret J. Kratina[1] | **Dennis R. DeVries**[1] | **Russell A. Wright**[1] | **Eric Peatman**[1] | **Steven J. Rider**[2] | **Honggang Zhao**[1]

[1]School of Fisheries, Aquaculture and Aquatic Sciences, Auburn University, Auburn, Alabama, USA

[2]Alabama Division of Wildlife and Freshwater Fisheries, River and Stream Fisheries Program, Montgomery, Alabama, USA

Correspondence
Garret J. Kratina
Email: gkratina@pa.gov

Present address
Garret J. Kratina, Pennsylvania Fish and Boat Commission, Division of Environmental Services, Bellefonte, Pennsylvania, USA

Honggang Zhao, Department of Natural Resources and the Environment, Cornell University, Ithaca, New York, USA

Abstract

Objective: We used two approaches, fish hard-part microchemistry and genetics, to quantify effects of low-use lock-and-dam structures on riverine fish movement. Each approach varied in temporal scope, with microchemistry addressing effects within a lifetime and genetics addressing effects across generations.

Methods: Water samples and individuals of two species (Paddlefish *Polyodon spathula* and Smallmouth Buffalo *Ictiobus bubalus*) were collected from four river sections that were separated by three low-use lock-and-dam structures on the Alabama River. Quarterly water samples were collected from 15 sites during 2017–2018, and concentrations of Sr, Ba, Mn, Mg, and Ca were quantified using mass spectrometry.

Result: Water elemental signatures were spatially variable but temporally consistent. The Sr:Ca ratios in fish hard parts differed significantly among river sections for both species. Additionally, discriminant function analyses classified fish to their river capture section with accuracy between 55% and 74% for Paddlefish (errors nearly always assigned individuals to adjacent river sections) and 37–47% for Smallmouth Buffalo. Population genetic analyses included fish from each river section, as well as from Alabama River tributaries and a neighboring watershed. Genotyping-by-sequence techniques identified 1,889 and 3,737 single nucleotide polymorphisms postfiltering in Paddlefish and Smallmouth Buffalo, respectively, which we used to estimate population diversity indices and conduct differentiation analyses. Analysis of molecular variance, discriminant analysis of principal components, Bayesian clustering, and pairwise comparisons of F_{ST} values indicated no strong evidence for genetic divergence in either species among river sections.

Conclusion: Within-lifespan results based on hard-part microchemistry suggested a potential for population isolation. However, longer-term genetic effects were not apparent, possibly because the life span of these large and relatively long-lived species means that few generations have passed since dam construction, and there could be sufficient mixing or population connectivity to prevent genetic divergence across river sections, particularly at the most downstream structure.

KEYWORDS

dentary bone microchemistry, genetics, otolith microchemistry, Paddlefish, Smallmouth Buffalo

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INTRODUCTION

Construction of dams in the United States proliferated during the 1930s, and while the rate of dam construction has decreased, effects due to substantial alteration and fragmentation of aquatic habitats have become apparent (Neves et al. 1997; Nilsson et al. 2005; Dudgeon et al. 2006). The fragmentation of lotic habitats by dams can obstruct migration and dispersal of resident and migratory species with the potential to reduce population persistence and recolonization (Fagan 2002; Geist 2011). This can also impact completion of species life cycles (Auer 1996; Joy and Death 2001; Cumming 2004), particularly those relying on natural flow regimes and extensive migration routes. Such impacts can potentially result in reduced genetic diversity and even species extirpation (Freeman et al. 2003; Faulks et al. 2011; Braulik et al. 2014).

To mitigate some effects of dams on migratory fish populations, fish passage operations have been implemented throughout the United States to help facilitate movements past dams and restore connectivity in riverine systems (Clay 1994; Jackson et al. 2001; Roscoe and Hinch 2010). Fish passage efforts in the USA have been concentrated on the Atlantic and Pacific coasts for anadromous species such as salmonids, Striped Bass *Morone saxatilis*, and clupeids (Jackson et al. 2001; Bernhardt et al. 2005). The southeastern USA has become a focus for studying the impacts of dams on fish populations in recent years due to this region's high density of dams (Graf 1999), combined with a number of endemic fishes and a large portion of its fauna being threatened (Warren et al. 2000). Specialized retrofits of existing dams to include fish passage facilities require substantial funding for planning, construction, and operation (Clay 1994). Therefore, it is critical to quantify the effects that dams have on fishes to determine the costs and benefits before implementing fish passage facilities.

The Alabama River watershed is an example of a southeastern U.S. ecosystem that has been influenced by the presence of dams. Dams on the Alabama River have modified flow and have led to fish assemblage simplification and species losses, particularly those that are diadromous and highly migratory (Freeman et al. 2005). As a result of these declines, understanding the degree to which fish passage is occurring and identifying opportunities for improving fish passage at dam structures on the Alabama River is important (Mettee et al. 2009, 2015; Simcox et al. 2015). For example, during normal flow conditions, the hydraulic turbulence through spill gates at each of the Alabama River dams restricts upstream movement of most fishes (Mettee et al. 2009, 2015). However, successful fish passage has been documented at one of the Alabama River dams during periods when flooding inundates an existing crested spillway (Mettee et al. 2009; Simcox et al. 2015). Specialized lock operations (termed "conservation lockages") during migration periods may help facilitate fish passage in some river systems or with some species but may not be effective in all situations (Mettee et al. 2009; Smith and Hightower 2012; Young et al. 2012). For example, while conservation lockages did increase the opportunity for passage and facilitated some passage events at two Alabama River dams, relatively few fish successfully passed upstream (Simcox et al. 2015).

Impact Statement

Paddlefish and Smallmouth Buffalo showed evidence of population separation by Alabama River lock-and-dams via a within-generation measure (chemical composition of fish jaw bones and ear stones) but not with a longer-term metric (genetic isolation). Both species are long lived relative to the age of the dams; effects may be more apparent in shorter-lived species.

Given the limited opportunities for fish passage in the Alabama River, here we quantify the influence of these dam structures on fish at a population level. Towards this end, we used two approaches, (1) hard-part microchemistry and (2) genetics, and two fish species that exhibit varying life history and migratory patterns, Paddlefish *Polyodon spathula* and Smallmouth Buffalo *Ictiobus bubalus*. These approaches were selected to evaluate potential population impacts at differing time scales, with hard-part microchemistry reflecting shorter-term impacts and genetic responses reflecting longer-term impacts. For example, while genetics can be useful for quantifying natal homing and stock structure or determining individual metrics such as paternity or genotype, it provides limited insight into short-term (within lifetime) individual movements, which is insight that can be gained through analysis of hard-part microchemistry. Investigating populations using a combination of techniques can help to increase understanding and inferential power (Pracheil et al. 2014).

Our objectives were to (1) quantify spatial and temporal variation in trace element signatures across the Alabama River and (2) use hard-part microchemistry and genetic analyses (SNP technology) to quantify potential fish population differences and the extent of mixing and connectivity of two fish species across distinct sections of the Alabama River that are separated by lock-and-dam structures as well as two upper tributaries.

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METHODS

Study site

The Alabama River is formed by the confluence of the Tallapoosa and Coosa rivers and flows for approximately 500 km across the coastal plain through central and southern Alabama before it joins the Tombigbee River, forming the Mobile River, which then flows into Mobile Bay (Figure 1). The headwaters of the Alabama River, the Coosa and Tallapoosa rivers, have wide ranging physiographic diversity, originating in the Blue Ridge, Valley and Ridge, and Upland Piedmont physiographic regions along the southern bend of Appalachia (Freeman et al. 2005).

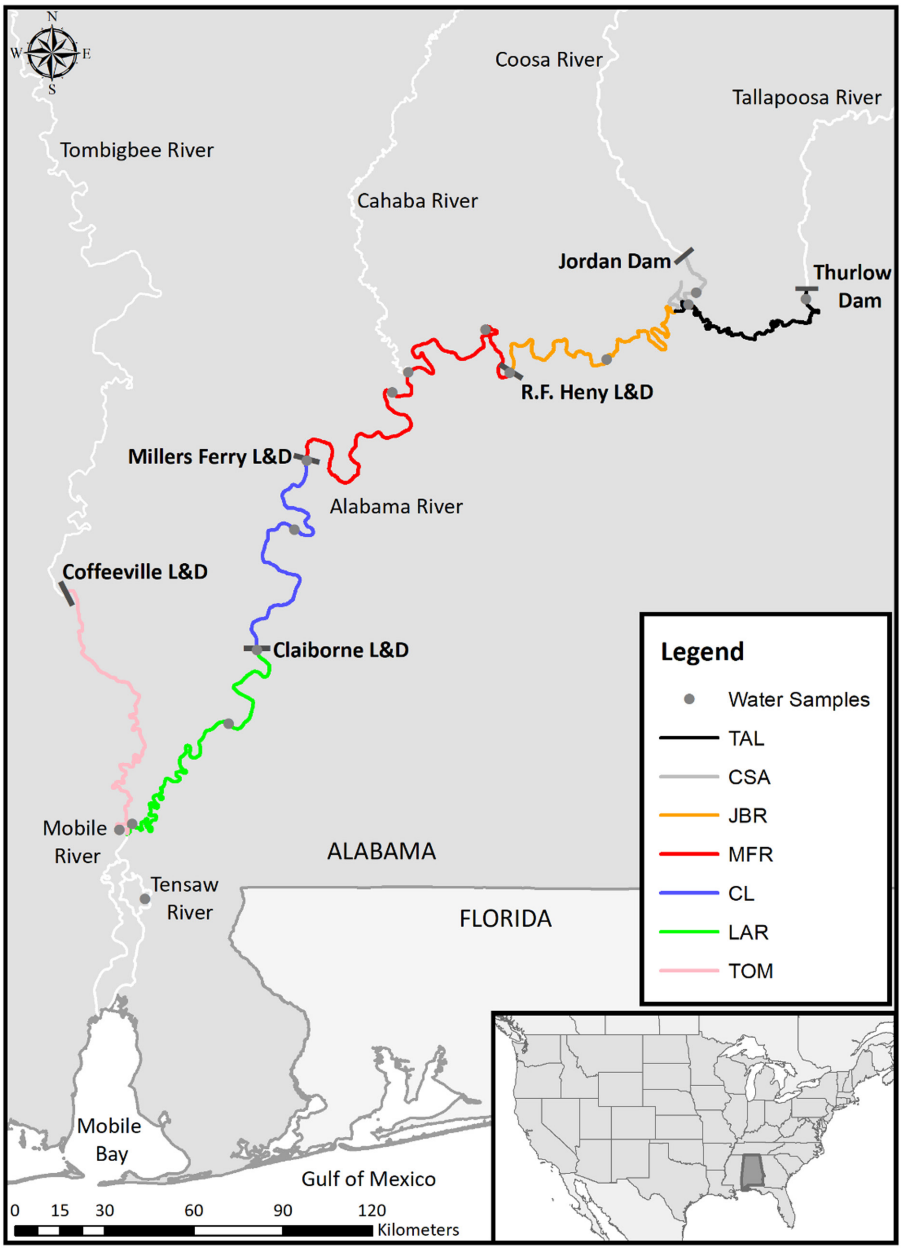

FIGURE 1 Map indicating the location of the Alabama River, its major tributaries, the three Alabama River lock-and-dam (L&D) structures, 15 water sampling locations, and four river sections, two tributaries, and a neighboring watershed where fish were collected (all indicated by unique colors). River sections are defined as follows: TOM = Tombigbee River (river kilometer [rkm] 0–188 on the Tombigbee River measuring from its confluence with the Alabama River), LAR = lower Alabama River (rkm 0–117 on the Alabama River measuring from its confluence with the Tombigbee River), CL = Claiborne Lake (rkm 117–214 on the Alabama River), MFR = Millers Ferry Reservoir (rkm 214–380 on the Alabama River), JBR = Jones Bluff Reservoir (rkm 380–491 on the Alabama River), CSA = Coosa River (rkm 0–30 on the Coosa River measuring from its confluence with the Tallapoosa River), and TAL = Tallapoosa River (rkm 0–80 on the Tallapoosa River measuring from its confluence with the Coosa River).

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The distinctive characteristics of each of these rivers are attributed to the varied lithography and soil horizons that occur in their watersheds (Freeman et al. 2005).

The United States Army Corps of Engineers Mobile District operates three low-use lock-and-dam structures on the Alabama River (Figure 1). From downstream to upstream these structures are Claiborne Lock-and-Dam (river kilometer [rkm] 117, measuring from the confluence of the Alabama River with Tombigbee River), Millers Ferry Lock-and-Dam (rkm 214), and Robert F. Henry Lock-and-Dam (rkm 380). These dams were completed between 1968 and 1971, and each contains gated spillways. Claiborne Lock-and-Dam differs from the others in that it does not have a hydropower structure and has both an ungated crested spillway that allows water flow at low to moderate flow conditions as well as gated spillways that are opened during high-water events. Laminar flow over the ungated spillway occurs when water levels reach an elevation of 10 m (U.S. Army Corps of Engineers 2013; Mettee et al. 2015).

Water sample collection

Water samples were collected quarterly (once during each season) between spring 2017 through summer 2018 from 15 sites (Figure 1). Sampling locations were selected to include a full spatial distribution throughout the Alabama River and its upper tributaries. A Van Dorn sampler was held open at a 1-m depth for 30 s to flush any residual water prior to sample collection (Farmer et al. 2013). The water sample was collected with a 50-mL sterile syringe from the Van Dorn sampler and filtered through a disposable 0.45-μm PTFE glass-fiber filter (Whatman GD/XP) directly into a 125-mL pre-acid-washed high-density polyethylene (HDPE) bottle containing 0.5 mL of 1:1 (≈34%) nitric acid solution for field preservation and storage. This process was repeated three times to fill each 125-mL HDPE bottle. Samples were stored on ice in the field, returned to the lab, and refrigerated until analysis (U.S. Environmental Protection Agency 1996; Lowe et al. 2011; Farmer et al. 2013).

Fish collection

Paddlefish were collected during 2012–2015 and 2017–2019, and Smallmouth Buffalo were collected during 2017–2018. Fish were captured using boat-mounted pulsed-DC electrofishing (DC GPP 7.5 electrofisher; Smith-Root) and gill nets (45.72 m in length, 152.4-mm bar measure) throughout the Alabama River. Additionally, Paddlefish samples were collected during the 2017 regulated commercial Paddlefish season occurring on the Alabama River, where commercial fishers used gill nets in designated stretches of the river. Harvested Paddlefish were checked by Alabama Division of Wildlife and Freshwater Fisheries biologists, who then collected biological data, fin clips, and dentary bones. Commercially harvested Paddlefish were required to be egg-bearing females with an eye-to-fork length >863.6 mm when measured with a flexible tape measure along the curvature of the body (Alabama Division of Wildlife and Freshwater Fisheries 2013). Location of capture assigned each fish to a population group, defined to be specific river sections separated by the three low-use lock-and-dam structures on the Alabama River, as well as its major tributaries, and a neighboring watershed (Figure 1). All fish were weighed (nearest 0.1 kg) and measured (nearest millimeter eye-to-fork length for Paddlefish, nearest millimeter TL for Smallmouth Buffalo).

Hard-part and tissue sample collection

Dentary bones were removed from each collected Paddlefish. Any flesh adhering to the extracted piece of dentary was removed, and dentary bones were allowed to air dry prior to sectioning (Scarnecchia et al. 2006; Bock et al. 2016). Both sagittal otoliths were removed from Smallmouth Buffalo with Teflon-coated forceps and cleaned for 30 s in 30% H_2O_2. Following cleaning, otoliths were rinsed in ultrapure water, dried, and stored in individual sterile polyethylene vials. Final sample sizes for hard-part microchemistry analysis included 186 Paddlefish dentary bones (30–56 per Alabama River section) and 209 Smallmouth Buffalo otoliths (42–53 per Alabama River section). Fin clips were collected from Paddlefish (left pectoral) and Smallmouth Buffalo (right pelvic) and preserved in 95% ethanol. Clips from 166 Paddlefish and 121 Smallmouth Buffalo were used for genotyping-by-sequencing (GBS) library construction and sequencing and those from 159 Paddlefish and 119 Smallmouth Buffalo were used for population genetic analyses.

Hard-part sample preparation

For Smallmouth Buffalo, one otolith from each fish was selected at random and set in an epoxy resin, and an approximately 0.5-mm transverse section through the core was removed using a low-speed diamond blade saw. Similarly, for Paddlefish a 0.50–0.65-mm transverse section was cut from the right-side dentary bone approximately 10 mm posterior to the point of greatest curvature (Scarnecchia et al. 1996; Bock et al. 2016). All sections were then polished sequentially with 300–14,000-grit

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lapping film until sections were smooth, after which they were rinsed with ultrapure water. Individual otolith sections were sonicated for 10 min, washed thoroughly once more with ultrapure water, and allowed to dry. Following drying, each section was mounted onto standard glass slides with cyanoacrylate glue (Ludsin et al. 2006; Farmer et al. 2013; Nelson et al. 2015; Daugherty et al. 2017). Slides were stored in sealed precleaned HDPE containers until microchemistry analysis and age estimation.

Hard-part microchemistry analysis

Hard-part microchemical analyses were conducted at Dauphin Island Sea Laboratory, Dauphin Island, Alabama, using laser ablation inductively coupled plasma mass spectrometry (LA-ICPMS) on a Agilent Technologies 7700 series ICPMS coupled to a 213-nm New Wave NWR-213 internally homogenized, optically attenuated laser system (Zeigler and Whitledge 2011; Bock et al. 2016; Nelson and Powers 2019).

Each dentary bone section had a linear transect ablated from inside its first annulus to the edge of its mesial arm, and each otolith section had a linear transect ablated from the focus of its core to its proximal edge. For all transects, a preablation pass was performed with 20% energy, 5 repetition rate (Hz), 40 spot size (μm), and 100 scan speed (μm/s). Laser parameters for each ablation were 25–28% energy, 10 repetition rate (Hz), 25 spot size (μm), and 5 scan speed (μm/s), quantifying the concentration (counts per second) of Ca^{43}, Sr^{88}, Ba^{137}, Mg^{24}, and Mn^{55}.

Before each ablation, 60 s of background signal were collected and duplicate runs of certified reference materials (CRMs) NIST-612, MACS-3, and MAPS-4 were performed before each day's analysis and after each subsequent hour of analysis. Limits of detection (3 SDs above background), background signal removal, instrument drift correction, and raw count transformation to concentration (ppm) were conducted using the Trace Elements IS Data Reduction scheme in Iolite version 3.63 built under IGOR Pro 7 software from WaveMetrics (2017), with Ca^{43} used as an internal standard held constant at 37.69% for otoliths with MACS-3 as the CRM and 27.00% for dentary bones with MAPS-4 as the CRM (Longerich et al. 1996; Ludsin et al. 2006; Paton et al. 2011; Nelson and Powers 2019). Concentrations (ppm) of Sr^{88}, Ba^{137}, Mg^{24}, and Mn^{55} obtained from the Iolite software were converted to molar ratios relative to Ca (μmol/mol).

All elements examined with LA-ICPMS (Ca, Sr, Ba, Mg, Mn) met the criteria (> limit of detection in 50% of samples) to be used in dentary bone (Paddlefish) and otolith (Smallmouth Buffalo) comparisons across all river sections in this study.

Water analysis

Dissolved elemental concentrations (Ca^{43}, Sr^{88}, Ba^{137}, Mg^{24}, and Mn^{55}) in water were quantified on the same Agilent ICPMS in solution mode at Dauphin Island Sea Laboratory. Water samples underwent a 10-fold dilution with a solution of 2% HNO_3. Additionally, each sample was spiked with internal standards Be^9 and In^{115} at concentrations of 10 and 1 ppm, respectively, to correct for instrument mass bias, drift, and matrix effects. An external five-point calibration curve with all elements of interest, and the same internal standards, was run before analysis began and used to convert raw count data into elemental concentrations (ppb), using Agilent Masshunter software. The NIST 1643f CRM also underwent a 10-fold dilution that was run every hour during analysis to assess instrument accuracy. The fourth point on the calibration curve and 2% HNO_3 blanks were also run every hour to further check for instrument drift and verify blank concentrations throughout analyses. All elemental concentrations output were converted to molar ratios relative to Ca (μmol/mol). All five elements (Ca, Sr, Ba, Mg, and Mn) analyzed met the inclusion criteria (i.e., being above limit of detection in >50% of samples) in the otolith analysis and were therefore quantified in our water samples.

Genetics

Genomic DNA from Paddlefish and Smallmouth Buffalo was extracted from fin clips using the DNeasy Blood & Tissue kit (Qiagen) according to the manufacturer's protocol. The DNA quality was assessed by running 100 ng of each DNA sample on 1% agarose gels. The DNA concentration was determined using the Quant-iT PicoGreen dsDNA Assay Kit (Invitrogen). All DNA samples were sent to the University of Minnesota Genomics Center for GBS library construction and sequencing. Briefly, 100 ng of DNA was digested with 10 units BamHI-HF and NsiI-HF (New England Biolabs) enzyme combination and incubated at 37°C for 2 h. Following digestion, samples were then ligated with 200 units of T4 ligase (New England Biolabs) and phased adaptors at 22°C for 2 h to inactivate the T4 ligase. The ligated samples were purified with SPRI (solid-phase reversible immobilization) beads and then amplified for 18 cycles with 2× HiFi Master Mix (New England Biolabs) to add barcodes. Libraries of Paddlefish and Smallmouth Buffalo samples were

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purified, quantified, pooled, and size selected for the 300- to 744-bp library region and diluted to 1.7 pm for sequencing. The pooled libraries were loaded across four lanes of a 150-bp single-read sequencing run on the NextSeq 550 instrument.

To perform reference-based SNP calling, we assembled the rough draft genomes for Paddlefish and Smallmouth Buffalo. One DNA sample from each species (LAR_19 and SBF_20_8) was selected and sent to the University of Minnesota Genomics Center for library construction and sequencing. During library creation, 100 ng of DNA was fragmented to target a 350-bp insert length using Covaris ultrasonic shearing. The sheared DNA was then end-repaired and subjected to a bead-based size selection. After adaptor and index ligation, the library was amplified using eight cycles of PCR. The amplified libraries were sequenced across two lanes of HiSeq 2500 125-bp paired-end run. A total of 98.7 gigabases (Gb) were generated from library sequencing and assembled into a 1.34 Gb (N50 = 3.85 kb, 562,901 scaffolds) Paddlefish draft genome (~73.7× genome coverage) using MaSuRCA version 3.2.4 with default settings (Zimin et al. 2013). Similarly, a total of 73.9 Gb data were generated and assembled into a 2.27 Gb Smallmouth Buffalo draft genome (~32.6 × coverage) with an N50 of 1.63 kb and 1,731,047 scaffolds. Genomewide SNPs were called using TASSEL (Trait Analysis by aSSociation, Evolution and Linkage) 5.0 GBS pipeline V2 (Glaubitz et al. 2014) with default parameter settings. During initial SNP filtering, the minimum minor allele frequency was set to 0.05 (overall), the minimum minor allele count was set to 10 (overall), and minimum locus coverage was set to 0.1 (overall). Only biallelic SNP were retained after initial filtering.

To ensure that the SNPs were polymorphic and informative for Paddlefish and Smallmouth Buffalo population genetics analyses, VCFtools (Danecek et al. 2011), TASSEL (Glaubitz et al. 2014), and GENEPOP (Rousset 2008) were used for stringent filtration of SNPs based on the following criteria: (1) individuals with more than 30% missing genotypes were removed, (2) the SNPs were called in at least 95% of individuals, (3) adjacent SNPs within 64 bp were removed to avoid loci that generated from ambiguous alignments, (4) SNPs deviating from Hardy–Weinberg equilibrium with p-value < 0.01 were removed, and (5) SNPs that showed significant linkage disequilibrium ($r^2 > 0.3$) and false discovery rate (p-value < 0.01) were removed.

Paralogous loci that arise during ancestry whole-genome duplication, autopolyploidy or allopolyploidy, or chromosomal duplication have been widely reported in species of plants, fish, and amphibians (Devos et al. 2005; Lien et al. 2016; Knytl et al. 2017). Such duplication events can facilitate species evolution and adaption but may lead to complicated genetic patterns where different ploidy levels exist in the genome (McKinney et al. 2018). Excluding paralogous loci is commonplace for GBS studies, regardless of their important role in species adaption and evolution (McKinney et al. 2017). Both species in this study were previously documented to have undergone whole genome duplication followed by genomic and chromosomal reorganization (Uyeno and Smith 1972; Ferris and Whitt 1977; Clements et al. 2004; Crow et al. 2012; Symonová et al. 2017). In order to keep only biallelic SNPs for downstream population genetics analyses, methods proposed by McKinney et al. (2017) were followed to identify paralogs from GBS data using two characteristics: the relative proportion of heterozygotes in a population (H) and the deviation of allele-specific reads of each locus from a 1:1 ratio (D). The SNP data sets after stringent filtering were used as input. The H and D for each locus were plotted to visualize the distribution of these variables across all loci using HDplot (with minor modification; https://github.com/hzz0024/paralog-finder). Based on the distribution plots, loci with $H > 0.5$ and $|D| > 4$ were classified as paralogs and excluded. The retained loci were used for population genetic analyses.

Genotyping and SNP discovery

For Paddlefish, a total of 152,964,388 high-quality reads were generated from Illumina NextSeq sequencing, with an average of 921,473 reads for each sequenced sample. A total of 99,086,273 high-quality reads were generated for Smallmouth Buffalo, with an average of 839,714 reads for each sequenced sample. During GBS analyses, TASSEL generated 18,137 and 56,744 prefiltered SNP loci for Paddlefish and Smallmouth Buffalo, respectively. Further individual filtering using 30% missing loci as a parameter reduced the sample size from 166 to 159 for Paddlefish and from 121 to 118 for Smallmouth Buffalo. To ensure that SNP data sets were polymorphic and informative for population genetics analyses, further stringent filtering steps using VCFtools, TASSEL, and GENEPOP software resulted in the identification of 2,770 and 4,248 SNPs for Paddlefish and Smallmouth Buffalo, respectively.

Paralog identification and marker characteristics

To keep only biallelic SNPs for downstream population genetics analyses, paralogs were identified from GBS data sets. Diploid SNP identification based on the relative proportion of heterozygotes (H) and the read deviation (D) found 1,889 (68.19%) of 2,770 markers

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to be diploid loci in Paddlefish and 3,737 (87.97%) of 4,248 SNPs to be diploid SNPs in Smallmouth Buffalo. Plotting *H* and *D* revealed two distinct clouds of loci in Paddlefish (Kratina 2019). The SNP loci within the inner dense cloud concentrated at 0.09–0.5 for *H* and 0.0 for *D*, suggesting the distribution of singleton loci. This cloud was ringed by a second cloud representing 31.81% of loci with clear deviation from equal ratios of sequence reads, consistent with expectations for duplicate loci. The Smallmouth Buffalo loci clustered into a cloud at 0.06–0.5 for *H* and 0.0 for *D*, corresponding to singleton loci distribution, with 12.03% loci outside this central range (Kratina 2019).

Data processing and statistical analyses

R Statistical Software (version 4.1.1; R Core Team 2019) was used for all statistical analyses.

Water chemistry

Akaike information criterion (AIC) was used to compare linear models that included unique combinations of the explanatory variables (site, season, and Claiborne gauge height). Site × season and site × gauge height interactions were included in the models compared by AIC. The model with the lowest AIC score for each element best explained the data. The explanatory variables identified from the best-fit linear model were then incorporated into an analysis of variance (ANOVA) model for each element to investigate differences among sites and identify patterns throughout the study area. A site × gauge height interaction was identified in the best-fit linear model for Mn; however, this interaction was not significant in the Mn ANOVA model. The ANOVA models were tested for normality of the residuals using the Shapiro–Wilks test and for homogeneity of variance using Levene's test. Although some ANOVAs passed these tests, some did fail. After testing appropriate data transformations, it was clear that no single transformation would correct for these violations of assumptions. Given that ANOVA is robust to violations of normality assumptions and has greater power than its nonparametric counterparts (Kruskal–Wallace or rank-sum approaches), it was still used (Brownie and Boos 1994; Underwood 1997; Khan and Rayner 2003; Nelson et al. 2014, 2015, 2017), and all ANOVA tests were conducted with untransformed data. Lastly, to determine consistency in element water chemistry between sampling years, paired *t*-tests (paired by site) for each element were performed to facilitate a comparison of the samples collected in spring of 2017 versus 2018 and the summer of 2017 versus 2018.

Hard-part edge to water comparison

Least squares linear regression was used to relate mean river section fish hard-part edge element concentration to mean water concentrations collected in proximity and time to fish collection. Hard-part "edge" was defined as the mean concentration ratios of the outer 20 μm of a fish's hard part (Farmer et al. 2013; Bock et al. 2016). Fish hard parts used for these comparisons were from Paddlefish collected during the spring 2017 and Smallmouth Buffalo collected during spring and fall 2017. A mean water element concentration for each river section was generated as the average across all water samples collected within the same river section during the season of interest.

Hard-part microchemistry

An ANOVA was used to determine if elemental concentrations averaged across hard-part transect profile regions differed among river sections. Mean core (first 20 μm), edge (last 20 μm), and whole otolith elemental concentrations were analyzed to test for any differences across the four sections of the Alabama River and its tributaries (when applicable).

For those elements with significant ($\alpha = 0.05$) differences among sites for a given hard-part segment, Tukey's honestly significant difference pairwise tests were used to compare mean hard-part element concentrations among river sections to determine site differences. Although data were not always distributed normally or did not always have homogeneous variance, ANOVAs using untransformed data were still used, given that the ANOVA test is robust to violations in normality assumptions and has greater power than their nonparametric counterparts (Kruskal–Wallace or rank–sum approaches). This is particularly true when sample sizes are large, as is the case here (Brownie and Boos 1994; Underwood 1997; Khan and Rayner 2003; Nelson et al. 2014, 2015, 2017).

Additionally, discriminant function analyses (DFAs) were performed to see how accurately fish could be classified to their capture locations based on the multivariate signatures observed in their hard-part laser ablation transects. The DFAs were performed for each region of the hard-part transect profile and included each element analyzed. An additional DFA performed combined lower Alabama River and Claiborne Lake Paddlefish into a single

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group (due to Claiborne Lock-and-Dam likely experiencing some degree of fish passage over its crested spillway). This resulted in three river sections for this second DFA in order to determine if classification accuracy for Paddlefish could be further improved by treating the lower two river sections as one. Fish collected in the Tallapoosa and Coosa rivers were not included in the DFAs as they did not help improve classification accuracy.

Finally, all box plots throughout this manuscript used the default box plot code in the statistical analysis software R, whereby the interquartile range (IQR; boxplot length) = Q_3 (quartile 3 [75th percentile]) − Q_1 (quartile 1 [25th percentile]), and the band within the box is quartile 2 (median [50th quartile]). The upper whisker was = Q_3 + 1.5 × IQR, and the lower whisker = Q_1 − 1.5 × IQR.

Population genetic analyses

Population genetic diversity indices were evaluated by computing the observed (H_o) and expected heterozygosity (H_e) using the R package Arlequin version 3.5 (Excoffier and Lischer 2010). The inbreeding coefficient (F_{is}) for each river section was calculated as well. Effective population size (N_e) was estimated using the linkage disequilibrium method implemented in NeEstimator version 2.1 (Do et al. 2014).

Population connection(s) were visualized with a discriminant analysis of principal components (DAPC) using the Adegenet package in R (Jombart and Ahmed 2011). The DAPC was run on individuals grouped by major river sections. In each case, the optimal number of principal components was determined by alpha-score procedure with 20 repeated runs. Another program based on Bayesian clustering algorithms, STRUCTURE version 2.3.4 (Pritchard et al. 2000), was used for population structure analysis. The admixture model with correlated allele frequencies was applied for data analysis with a burn-in of 20,000 iterations followed by 200,000 Markov chain–Monte Carlo repetitions. Different numbers of assumed population genetic clusters (K = 1–10) were used to determine the best K value using the Web server STRUCTURE HARVESTER (Earl and VonHoldt 2012) and were repeated 10 times for each K value.

Population differentiation analyses were performed for all pairs of major river sections using analysis of molecular variance (AMOVA) and the Weir and Cockerham estimator of F_{ST} (Weir and Cockerham 1984) in Arlequin version 3.5 (Excoffier and Lischer 2010). Significance level for each F_{ST} comparison was estimated using 10,000 permutations with false discovery rate correction (initial $P = 0.05$). The significance level of the AMOVA test was also estimated using 10,000 permutations.

RESULTS

Water chemistry

All four element ratios exhibited significant variation across sites (Table 1; Figure 2). In addition, all but one comparison between years for spring or summer ratios of Sr:Ca, Ba:Ca, Mg:Ca, and Mn:Ca were not significant (t-tests: all p-values > 0.21; spring Mg:Ca ratios $p = 0.02$). See Supplemental Tables S1–S4 (available in the online version of this article) for the results of the AIC scores.

Correlation of water to hard-part edge

Paddlefish

Mean Paddlefish dentary bone edge element concentration was significantly related to mean water element concentrations across river sections for both Sr and Mn in spring 2017. Simple linear regressions between water and dentary edge chemistry were positive for Sr:Ca ($R^2 = 0.98$, $p = 0.008$) and Mn:Ca ($R^2 = 0.91$, $p = 0.044$). Linear regressions between water and dentary bone chemistry were not significant for Ba:Ca ($R^2 = 0.01$, $p = 0.891$) or Mg:Ca ($R^2 = 0.72$, $p = 0.154$). See Supplemental Figure S7 (available in the online version of this article) for these graphs.

Smallmouth Buffalo

Mean Smallmouth Buffalo otolith edge Sr:Ca ratio was significantly related to mean water Sr:Ca ratio across river sections in fall 2017 ($R^2 = 0.92$, $p = 0.043$) and marginally significantly ($0.05 < p < 0.10$) related in spring

TABLE 1 Results of one-way ANOVA models comparing water Sr:Ca, Ba:Ca, Mg:Ca, and Mn:Ca among water sampling sites, along with other explanatory variables (season and gauge height) identified via Akaike information criterion from the best-fit linear model for each element.

Signature	Comparison	*N*	*F*	df	*p*
Sr:Ca	Site	89	88.716	14	<0.01
	Season	4	2.954	3	0.04
	Gauge height	89	12.293	1	<0.01
Ba:Ca	Site	89	20.640	14	<0.01
	Gauge height	89	16.810	1	<0.01
Mg:Ca	Site	89	35.376	14	<0.01
	Gauge height	89	2.608	1	0.11
Mn:Ca	Site	89	3.358	14	<0.01
	Site : gauge height		0.308	14	0.99

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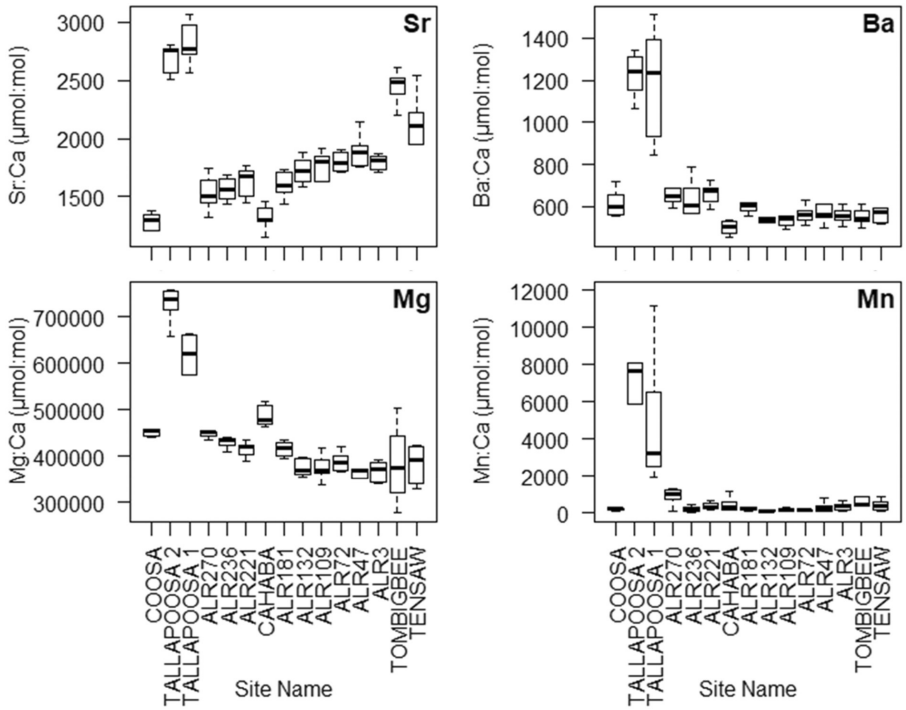

FIGURE 2 Trace element water sample concentration (Sr:Ca, Ba:Ca, Mg:Ca, and Mn:Ca) across sites ranging from upstream (to the left of the *x*-axis) to downstream (to the right of the *x*-axis). Each site is represented by a box plot, where the horizontal line in each box indicates the median, the box dimensions represent the 25th to 75th percentile ranges, and whiskers show the 10th to 90th percentile ranges. The numbers by Alabama River (ALR) site names indicate the river mile of that site.

2017 samples ($R^2=0.99$, $p=0.077$). Linear regressions between water and otolith chemistry in spring 2017 were not significant for Ba:Ca ($R^2=0.66$, $p=0.399$), Mg:Ca ($R^2=0.77$, $p=0.318$), or Mn:Ca ($R^2=0.01$, $p=0.952$). Simple linear regressions between water and otolith chemistry in fall 2017 were positive for Mn:Ca ($R^2=0.87$, $p=0.066$) but not for Ba:Ca ($R^2=0.12$, $p=0.653$) or Mg:Ca ($R^2=0.50$, $p=0.294$). See Supplemental Figure S8 for these graphs.

Hard-part microchemistry

Paddlefish comparison among river sections

Paddlefish dentary bone mean whole transect Sr:Ca ratios differed significantly among river sections (ANOVA: $F_{4,\ 181}=69.31$, $p<0.001$). Dentary bone mean edge Sr:Ca ratios also differed significantly among river sections (ANOVA: $F_{4,\ 181}=66.05$, $p<0.001$) as did dentary bone mean first 20 µm Sr:Ca ratios (ANOVA: $F_{4,\ 178}=41.16$, $p<0.001$). The Sr:Ca ratios were all highest in LAR (Figure 3; Kratina 2019). Paddlefish dentary bone Ba:Ca ratios also differed across sites for whole transect, first 20 µm, and edge (all $p \leq 0.001$), being highest at the Tallapoosa River site and slightly decreasing with progressive distance downstream (Supplemental Figure S1; Kratina 2019). And both Mg:Ca and Mn:Ca ratios differed across river locations for whole transect, first 20 µm, and edge dentary bone measures (all $p \leq 0.004$; Supplemental Figures S2 and S3; Kratina 2019).

Paddlefish discrimination among river sections

A first set of discriminant function analyses assigned fish to the correct river collection section with a moderate degree of accuracy (55–74%) across dentary regions (Table 2). Fish collected in MFR consistently had the highest classification percentages for each region of the dentary. Elements that contributed to the discriminating axes were Ba^{137}, Sr^{88}, and Mn^{55} (in decreasing order of importance). The relative similarity in concentration of these elements in adjacent river sections limited the ability to discriminate between these areas and misclassifications almost always occurred to an adjacent river section.

Smallmouth Buffalo comparison among river sections

Smallmouth Buffalo otolith mean whole transect Sr:Ca ratios differed significantly among river sections (ANOVA:

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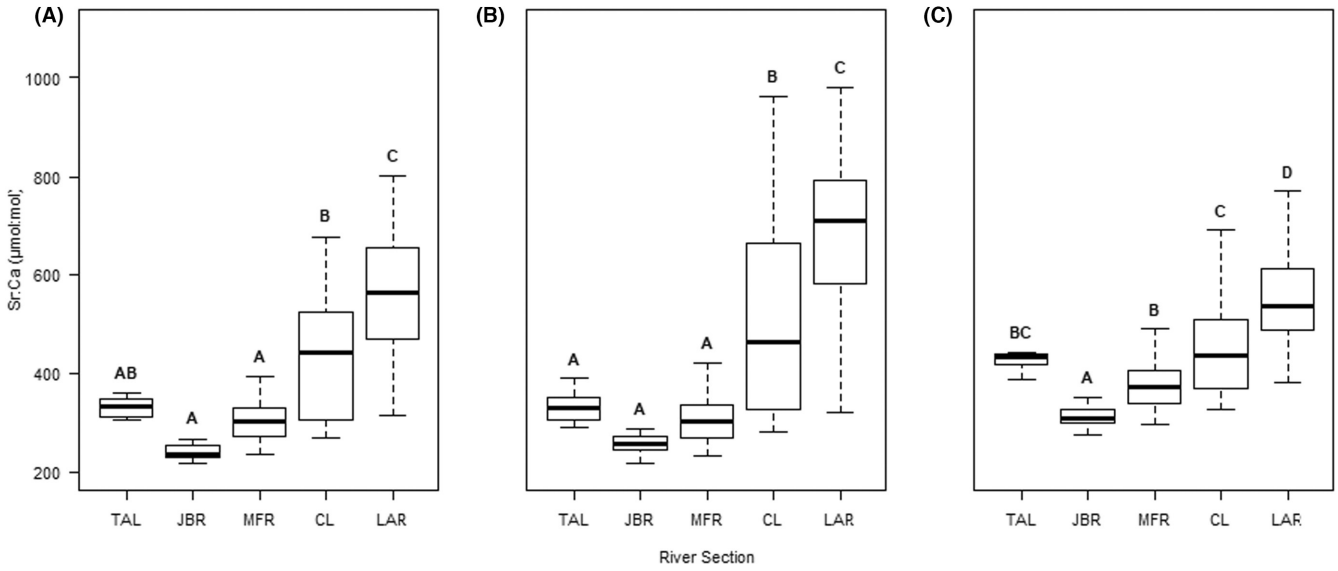

FIGURE 3 Concentrations of Sr:Ca in Paddlefish dentary bones for each region of the ablation transect analyzed, showing (A) mean transect, (B) mean edge, and (C) mean first 20 µm. Each river section has a box plot that represents the observed values, and different letters above the box plots indicate significant differences among river sections based on Tukey pairwise comparisons. For the box plots, the horizontal line in each box indicates the median, the box dimensions represent the 25th to 75th percentile ranges, and whiskers show the 10th to 90th percentile ranges. See Figure 1 for river section definitions.

$F_{5,\,203} = 3.142$, $p = 0.009$), being lowest in JBR (Figure 4A). Otolith mean edge Sr:Ca ratios differed significantly among river sections (ANOVA: $F_{5,\,203} = 4.714$, $p < 0.001$), being highest in TAL (Figure 4B). Otolith first mean 20 µm Sr:Ca ratios did not differ among river sections (ANOVA: $F_{5,\,203} = 1.941$, $p = 0.089$) (Figure 4C). Smallmouth Buffalo whole transect, edge, and first 20 µm Ba:Ca ratios did not differ across river sections (all $p > 0.67$; Supplemental Figure S4). The Mg:Ca ratios differed across river locations for whole transect, first 20 µm, and edge measures (with most measures showing a slight increasing pattern with progressive distance downstream; all $p \leq 0.017$; Supplemental Figure S5; Kratina 2019), and Mn:Ca ratios for whole transect and edge differed across river sections (higher upstream; both $p < 0.001$), while that for first mean 20 µm did not ($p = 0.28$; Supplemental Figure S6).

Smallmouth Buffalo discrimination among river sections

Discriminant function analyses assigned fish to the correct river collection section with a low degree of accuracy (37–47%) in all otolith regions (Table 2). The river section of highest classification accuracy varied among otolith regions, with JBR being most accurate for the transect mean, MFR most accurate for the mean edge, and LAR being most accurate for the first 20 µm mean. Elements that contributed to the discriminating axes were Ba^{137}, Sr^{88}, and Mn^{55} (in decreasing order of importance). The relative similarity in concentration of these elements in adjacent river sections limited the ability to discriminate between these areas.

Genetics

Marker characteristics

Observed (H_o) and expected heterozygosity (H_e) were calculated for fish from each river section (Table 3). Heterozygosity rate gradually decreased downstream from JBR to LAR. Inbreeding coefficients (F_{is}) were calculated for fish collected from each river section, with highest levels of inbreeding observed in LAR for both species (Table 3).

Population structure and differentiation

The AMOVA for Paddlefish and Smallmouth Buffalo data sets did not detect genetic differentiation among river sections, and most genetic variation was generated within individuals (Table 4). In Paddlefish, 96.16% of the total variation was explained by differences within individuals, 3.84% was due to variation among individuals within populations, and variation among populations did not contribute to the total variation. In Smallmouth Buffalo, 93.68% of the total variation was explained by differences within individuals, 6.28% was due to the

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TABLE 2 Results of the discriminant function analysis for Paddlefish (PAD) and Smallmouth Buffalo (SBF) using river section as each group in the analysis. Discriminant function analyses were performed on each part of the ablation transect. See Figure 1 for river section definitions.

Transect analysis	Species	Collection location	Assigned collection location				% Correct	Total classification accuracy (%)
			JBR	MFR	CL	LAR		
Mean transect	PAD	JBR	20	10	0	0	67	68
		MFR	1	47	1	1	94	
		CL	1	14	16	12	37	
		LAR	1	8	8	39	70	
	SBF	JBR	26	8	1	7	62	43
		MFR	3	26	9	10	54	
		CL	5	16	23	9	43	
		LAR	6	16	21	9	17	
Mean edge	PAD	JBR	24	6	0	0	80	74
		MFR	0	47	3	0	94	
		CL	1	16	14	12	33	
		LAR	1	3	5	47	84	
	SBF	JBR	23	12	0	7	55	47
		MFR	3	29	5	11	60	
		CL	9	15	14	15	26	
		LAR	2	18	7	25	48	
Mean first 20 μm	PAD	JBR	8	22	0	0	27	55
		MFR	2	39	5	4	78	
		CL	0	19	12	11	29	
		LAR	0	7	10	37	69	
	SBF	JBR	12	15	7	8	29	37
		MFR	5	22	14	7	46	
		CL	4	16	12	21	23	
		LAR	4	13	8	27	52	

variation among individuals within populations, and variation among populations only accounted 0.04% of the total variation.

To examine genetic differentiation among populations and their degree of variance in allele frequencies, pairwise F_{ST} values were measured (Table 5). The F_{ST} statistics in this study were, in general, exceptionally low, and only one pairwise comparison was significant (between JBR and MFR in Smallmouth Buffalo). This suggests that JBR and MFR are the most genetically divergent river sections for Smallmouth Buffalo. Meanwhile, no significant F_{ST} values were detected for Paddlefish between river sections. While F_{ST} values between CSA Paddlefish and other river sections were relatively higher than the other comparisons, this was most likely due to the small sample size ($n=4$) from the CSA.

We used STRUCTURE software to examine population structure for both species, examining the optimal number of upper genetic clusters (K) for the GBS data sets using the 1,889 SNPs for Paddlefish and 3,737 SNPs for Smallmouth Buffalo. For both species, the optimal numbers of uppermost genetic clusters (K) were 1 ($K=1$) according to the calculated probabilities of each K. This suggested a lack of spatial genetic clustering between fish in each river section.

Lastly, DAPC also showed a general pattern of low genetic differentiation, similar to that of the previous analyses performed. The DAPC plot of Paddlefish, including all river sections and SNPs, showed no clear separation or clusters among river sections (Figure 5), with only CSA samples slightly isolating from all other river sections. This pattern was consistent with the F_{ST} results, potentially due to the small sample size from the CSA. The DAPC analysis of Smallmouth Buffalo with all SNPs also showed no clear genetic clusters or subdivision among river sections (Figure 6).

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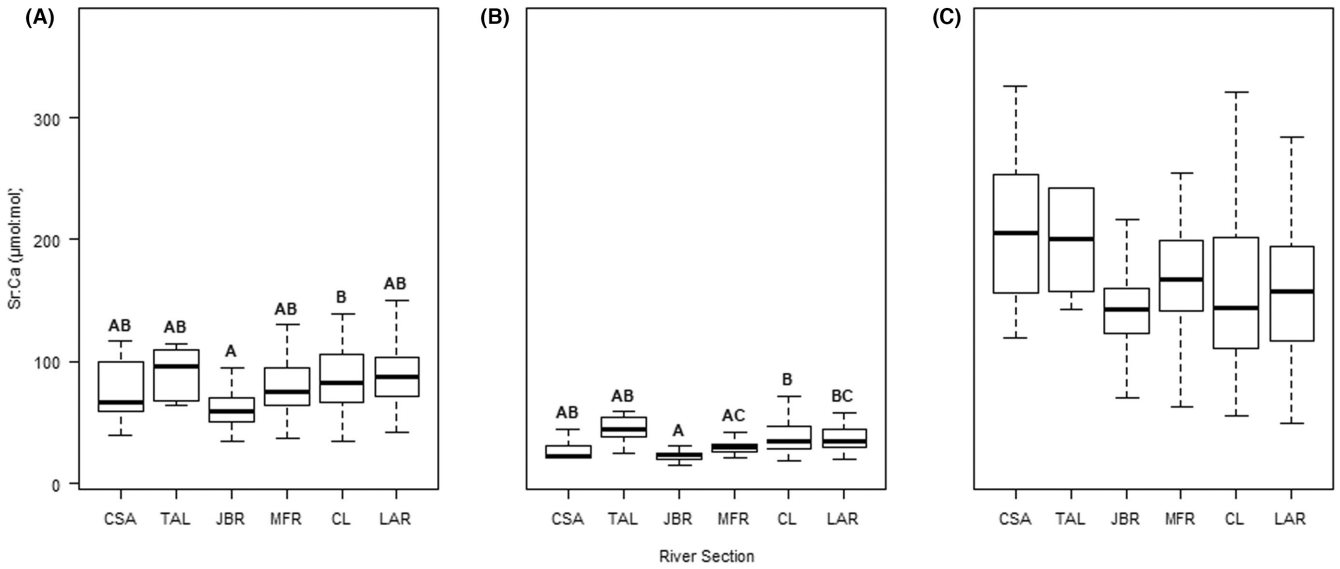

FIGURE 4 Concentrations of Sr:Ca in Smallmouth Buffalo otoliths for each region of the ablation transect analyzed, showing (A) meant transect, (B) mean edge, and (C) mean first 20 µm. Each river section has a box plot that represents the observed values, and different letters above the box plots indicate significant differences among river sections based on Tukey pairwise comparisons. For the box plots, the horizontal line in each box indicates the median, the box dimensions represent the 25th to 75th percentile ranges, and whiskers show the 10th to 90th percentile ranges. See Figure 1 for river section definitions.

TABLE 3 Genetic diversity parameters of Paddlefish and Smallmouth Buffalo from each river section. Abbreviations are as follows: H_o indicates average observed heterozygosity, H_e indicates average expected heterozygosity, F_{is} indicates inbreeding coefficient, and N_e indicates the estimate of contemporary effective population size (with 95% confidence intervals in parentheses). See Figure 1 for river section definitions.

Species	River section	Sample size	H_o	H_e	F_{is}	N_e
Paddlefish	CSA	4	0.389	0.403	0.034	
	JBR	27	0.291	0.304	0.041	844 (670–1,138)
	MFR	33	0.289	0.302	0.040	1,516 (1,124–2,324)
	CL	35	0.290	0.302	0.037	1,198 (954–1,607)
	LAR	31	0.286	0.301	0.045	1,960 (1,317–3,812)
	TOM	29	0.294	0.304	0.032	1,673 (1,144–3,099)
	Mean		0.306	0.319	0.038	1,438
Smallmouth Buffalo	JBR	29	0.238	0.250	0.043	1,739 (1,398–2,299)
	MFR	30	0.232	0.249	0.060	1,619 (1,303–2,134)
	CL	30	0.232	0.250	0.070	Infinite
	LAR	29	0.228	0.249	0.078	Infinite
	Mean		0.232	0.249	0.063	

DISCUSSION

We quantified short-term (hard-part microchemistry) and long-term (genotype by sequencing) effects of three Alabama River lock-and-dam structures on the population structure of two fish species. Previous work has used tagging and tracking of individuals to show that individual fish movement was restricted to varying degrees by the dams (Simcox et al. 2015; Hershey et al. 2022), and here we wanted to quantify potential effects at longer time scales (hard-part microchemistry = within the life span of an organism, genetics = across generations); however, it appears that longer-term genetic divergence was not manifesting in these long-lived species, perhaps in part because the dams have been in place for a relatively short time relative to the lifespans and potential generation times of these species. Below we further consider and discuss these findings.

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TABLE 4 Analysis of molecular variance for the SNP genotypes of Paddlefish and Smallmouth Buffalo. An asterisk indicates $p < 0.001$.

Species	Source of variation	df	Sum of squares	% Variation
Paddlefish	Among populations	4	1,669.19	0.02
	Among individuals within populations	150	61,687.62	7.22*
	Within individuals	155	55,329	92.76*
	Total	309	118,873.81	
Smallmouth Buffalo	Among populations	3	1,489.05	0.04
	Among individuals within populations	114	55,403.58	6.28*
	Within individuals	118	50,567.00	93.68*
	Total	235	107,459.63	

TABLE 5 Pairwise F_{ST} values between river sections for Paddlefish (below the diagonal) and Smallmouth Buffalo (above the diagonal). Bold italics indicates significance after false discovery rate correction set at $p < 0.05$. See Figure 1 for river section definitions.

River section	CSA	JBR	MFR	CL	LAR	TOM
CSA						
JBR	0.00322		***0.00232***	0.00051	0.00149	
MFR	0.00013	0.00056		0.00203	0.00206	
CL	0.00279	0.00127	0.00064		0.00045	
LAR	0.00397	−0.00066	0.00057	0.00109		
TOM	−0.00017	0.0005	−0.00043	0.00125	−0.00007	

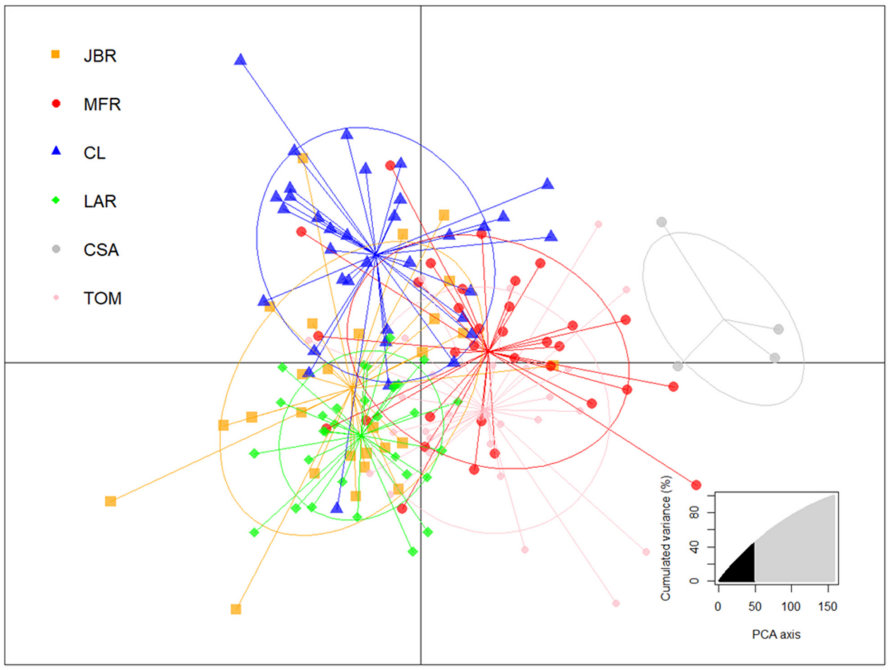

FIGURE 5 Scatterplot output from discriminant analysis of principal components for genetic signatures from SNPs of Paddlefish individuals (based on alpha score of 49). Colors indicate river sections as in Figure 1.

Water chemistry

Element : Ca ratios in water (particularly Sr:Ca) were spatially variable (and consistent with watershed geology; Newton et al. 1987; Wells et al. 2003) but temporally consistent, as needed for hard-part microchemistry studies (Campana 1999; Elsdon et al. 2008; Walther and Limburg 2012; Pracheil et al. 2014). In addition, strontium

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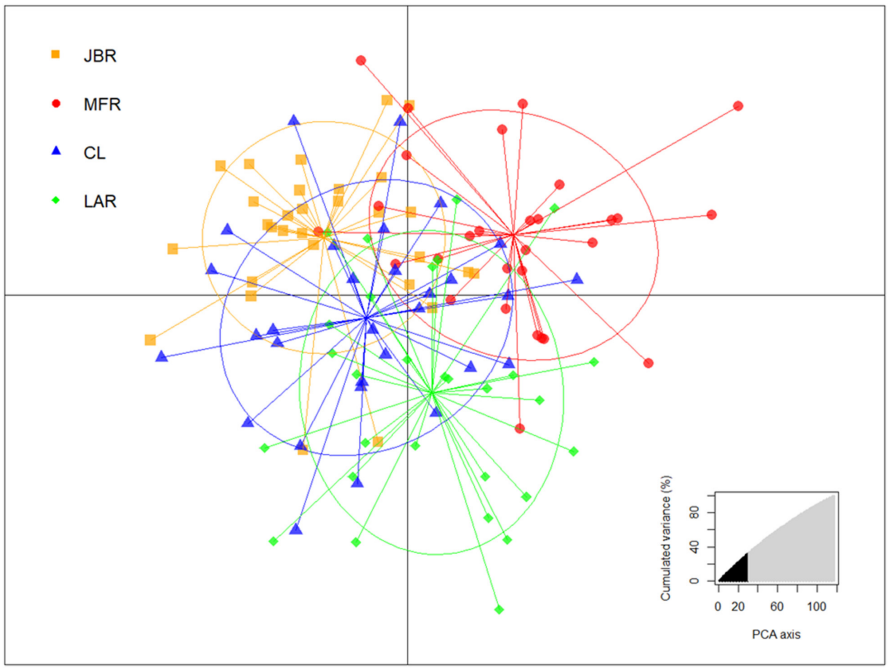

FIGURE 6 Scatterplot output from discriminant analysis of principal components for genetic signatures from SNPs of Smallmouth Buffalo individuals (based on alpha score of 30). Colors indicate river sections as in Figure 1.

concentration increased downstream in proximity to the coastal and estuarine environment (Elsdon et al. 2008; Walther and Limburg 2012; Farmer et al. 2013), with Sr:Ca ratio ranges consistent with previous freshwater studies (Kraus and Secor 2004; Farmer et al. 2013; Daugherty et al. 2017). While element : Ca ratios in large river systems, particularly those not connected to downstream marine habitats, can vary temporally (e.g., Phelps et al. 2012), our data over 2 years of sampling were temporally consistent (for the spring and summer comparisons) as has been found in some previous riverine work (Morissette and Sirois 2021). Given that our target species have life spans that exceed this 2-year period, their accumulated hard-part microchemistry likely includes periods outside our two water chemistry sample years, making the temporal stability in the water an important observation.

Among sites, the Tallapoosa River was unique with extremely high element : Ca ratios. This uniqueness derived from the geology of this system (Freeman et al. 2005) due to lower calcium concentrations, leading to increased trace element : Ca ratios. When absolute element concentrations (other than calcium) for this tributary were considered, they mimicked the north-to-south patterns observed for the rest of the watershed. The elevated element : Ca ratios in the Tallapoosa River proved to be particularly important when analyzing fish hard parts collected from the Tallapoosa River.

Hard-part microchemistry

The Sr:Ca ratios in ambient water were correlated with those in Paddlefish dentary bone and Smallmouth Buffalo otolith edge material, which is required for microchemistry studies to be effective in reconstructing environmental histories (Campana 1999; Pracheil et al. 2014). Dentary bone microchemistry proved particularly valuable for quantifying riverine Paddlefish population connectivity and natal origin as has been suggested previously (Gillanders and Kingsford 1996; Campana 1999; Campana et al. 2000; Thorrold et al. 2001; Gillanders 2002). Trace element concentrations compared across river sections and DFA classifications supported that some population groups are mixing, while others remain isolated, and strontium, barium, and manganese (based on their contribution to the discriminating axes) contributed to determining Paddlefish population connectivity, movement, and natal origin. Based on these results, fish in Millers Ferry Reservoir and Jones Bluff Reservoir appear functionally isolated from Paddlefish populations in Claiborne Lake and the lower Alabama River, with movement within these systems essentially blocked by Millers Ferry and Robert F. Henry dams (Thomas 2020).

We expected to see stronger differences in Smallmouth Buffalo otolith microchemistry across river sections given their reduced migratory nature. As with Paddlefish, strontium best described population connectivity or separation patterns for Smallmouth Buffalo, but patterns were not

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nearly as strong or distinct as in Paddlefish. The Sr:Ca concentration in otoliths increased with distance downstream, but only samples from river sections furthest apart differed significantly. The DFA classification accuracy for Smallmouth Buffalo was much lower versus that for Paddlefish.

The crested spillway that is periodically inundated during high-flow events at Claiborne Lock-and-Dam provides an opportunity for fish populations to mix across this barrier (as has been seen in this and other low-head dam systems; Nichols and Louder 1970; Smith and Hightower 2012; Simcox et al. 2015). Fish may also pass via lock chambers (Zigler et al. 2004; Tripp et al. 2014), although this behavior has been limited on the Alabama River (Simcox et al. 2015). Significant differences observed in mean Sr:Ca over multiple parts of dentary bone ablation transects, reflecting different times during a Paddlefish's life, suggest that the Alabama River sections are relatively isolated. However, when examining the DFA classifications for these lower sections, the results suggest the potential for some limited mixing as indicated by misclassifications among river sections as has been observed previously (Rooker et al. 2008; Geffen et al. 2011).

Relative to natal origins (as in Thorrold et al. 1998; Whitledge et al. 2007; Walther et al. 2008; Gahagan et al. 2012), we found element : Ca ratios in the first 20 μm to be clearly distinguishable across river sections, similar to results from the edge and whole transect, suggesting that Paddlefish reproduction is occurring within all river sections of the Alabama River. Paddlefish spawning has been observed to occur below dams and in sections of rivers separated by dams (Ruelle and Hudson 1977; Pasch et al. 1980; Wallus 1986), suggesting that the pattern being observed here occurs more broadly geographically. Interestingly, similar patterns were not seen in Smallmouth Buffalo otoliths.

Patterns in Sr:Ca across river sections observed in Smallmouth Buffalo versus Paddlefish may be due to several factors. First, these species exhibit different habitat use patterns. Paddlefish are pelagic (Clark-Kolaks et al. 2009) and occupy a variety of habitats, including backwaters, oxbows, and main-channel areas (Boschung and Mayden 2003). In contrast, Smallmouth Buffalo are more benthic (Gido 2002), although they also occupy a diversity of habitats, including main channels, backwaters, and other low-water-velocity habitats (Kallemeyn and Novotny 1977; Edwards and Twomey 1982). Backwaters and side-channel areas may have different trace element signatures compared with that of the main channel, which would not have been detected through our main-channel and tributary water sampling. Additionally, different hard-part structures were used for Smallmouth Buffalo versus Paddlefish, and different species may incorporate elements into their otoliths differently (Gillanders and Kingsford 2003; Rooker et al. 2004; Pracheil et al. 2014). Therefore, our results for Paddlefish versus Smallmouth Buffalo may have been influenced by a combination of being different species and use of different hard-part structures (dentary bones versus otoliths).

Relative to the upper tributaries, Paddlefish in the Tallapoosa and Coosa rivers (tributaries that form the Alabama River) are believed to move between Jones Bluff Reservoir and these tributaries (Lein and DeVries 1998; DeVries et al. 2009, S. J. Rider, Alabama Division of Wildlife and Freshwater Fisheries, personal observation). Water trace element : Ca ratios were significantly higher in the Tallapoosa River, allowing us to test this hypothesis. Significant differences in fish collected from TAL versus JBR were identified for Sr:Ca in the first 20 μm portions of the ablation transect. Higher Sr:Ca concentrations were observed for the whole transect and edge in TAL fish as well, but these were not significantly different from fish collected from JBR. This may indicate that Paddlefish collected in the Tallapoosa River move freely between Jones Bluff Reservoir and the Tallapoosa River, most likely for reproduction purposes. Additionally, significant differences observed in the natal (first 20 μm) region of ablation transects indicates spawning site fidelity as observed in other studies of Paddlefish (Lein and DeVries 1998; Stancill et al. 2002; Firehammer and Scarnecchia 2006; Jennings and Zigler 2009). In other words, Paddlefish collected in the Tallapoosa River appear to return to and reproduce in this system, whereas fish collected from Jones Bluff Reservoir do not move upstream into the Tallapoosa River to spawn (similar to individual variation in behavior seen in salmonids; Birnie-Gauvin et al. 2021).

Genetics

Genotyping-by-sequencing techniques proved particularly valuable here, identifying thousands of SNPs for each species for population genetic analyses. Acipenseriform fishes share a tetraploid ancestor followed by genomic and chromosomal reorganization, as do Catostomidae species (Uyeno and Smith 1972; Schwemm et al. 2014). Therefore, it was critical to identify diploid markers rather than polypoid markers because characteristics such as ambiguous homology and dosage uncertainty typically prevent the use of polyploid markers for population genetic analyses. When identifying biallelic SNPs via HDplot, Paddlefish showed a similar distribution of z-score (D) and proportion of heterozygotes (H) as did mountain barberry *Berberis alpine* (McKinney et al. 2017), a shrub also known to have a large proportion of duplicate loci (35%), which was confirmed by the same paralog identification method

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(Mastretta-Yanes et al. 2014; McKinney et al. 2017). Additionally, previous efforts in Paddlefish microsatellite development have shown varied proportions of singleton loci in marker discovery, ranging from 52% (13 of 25; Schwemm et al. 2014) to 75% (6 of 8; Heist et al. 2002), which overlapped the singleton loci rate (68.19%) in this study. Smallmouth Buffalo loci were primarily clustered into one cloud, corresponding to singleton loci distribution, with only a small percentage outside its central range. This suggests that duplicate loci were present at low frequency in the Smallmouth Buffalo genome, potentially due to extensive rediploidization after whole-genome duplication (Uyeno and Smith 1972). As a result of the effectiveness of this approach in selecting representative singleton SNPs, our optimized parameters for multiplex panel design and validation can benefit future studies.

This study also represents the first population genetics assessment of Paddlefish using SNP markers. Other Paddlefish population studies have been conducted using allozymes, mitochondrial DNA, or microsatellite markers (Carlson et al. 1982; Epifanio et al. 1996; Heist and Mustapha 2008; Sloss et al. 2009; Zheng et al. 2014). For example, a previous genetic survey using seven microsatellite loci observed higher genetic heterozygosity values in Paddlefish populations from Mississippi (0.68) and Poland stocks (0.59–0.60, imported from USA) (Kaczmarczyk et al. 2012) compared with this study with SNPs. Given that microsatellites typically have more alleles and higher heterozygosity than SNPs (Tokarska et al. 2009) and that evidence from other animal species have shown only weak or no congruence between the estimates of heterozygosity derived from the two marker types (Vignal et al. 2002), we did not directly compare heterozygosity. Additionally, our Paddlefish inbreeding levels were generally consistent with a previous study in four geographically separate Paddlefish populations, with F_{is} ranging from 0.013 (Tallapoosa River) to 0.099 (Yellowstone–Missouri River) (Zheng et al. 2014). Overall, heterozygosity values were neither very low nor high, suggesting that fish in each river section have an intermediate level of genetic variability. Observed heterozygosities were slightly lower than expected, which may be attributed to inbreeding. However, when looking at inbreeding coefficients (F_{is}) for each river section, they were low, indicating that inbreeding does not currently appear to be an issue for these species in any river section.

Also, estimates of effective population sizes of these species for each river section, particularly for Paddlefish, can provide managers with critical information for best management of these species. The metric N_e is a commonly used population estimation measurement as it only requires a single population for calculation (see equation in Cervantes et al. 2011). Some factors such as the variance in reproductive success of individuals (Heist and Mustapha 2008), sample size (normally <1% of the census population), and overlapping generations may impact the accuracy of N_e and generally lead to underestimate of population size (Marandel et al. 2019). For example, a simulation study on Thornback Ray *Raja clavata* found NeEstimator underestimated N_e by 31% (Marandel et al. 2019), while Waples et al. (2014) found a 10% bias for Atlantic Cod *Gadus morhua*. Therefore, particular attention should be paid to the interpretation of N_e results. Infinite values of N_e for Smallmouth Buffalo were potentially due to the limited sample size leading to overestimation (Marandel et al. 2019). Small sample sizes may also affect the accuracy of our Paddlefish N_e estimate, and they also may be biased due to overlapping generations and/or Paddlefish life history traits (i.e., iteroparity, delayed maturity, and high fecundity; Epifanio et al. 1996; Heist and Mustapha 2008).

Population structure and differentiation analyses all converged on a similar conclusion that there was no strong evidence for population divergence among river sections for either species. When initially designing this study, it was believed that if there would be genetic differentiation among river sections, it would be observed between those that were separated by the greatest distance, and that it would be more prevalent in a species with less migratory capabilities. Almost no genetic differentiation was observed between any river sections for either species. The only differentiation observed did occur for the less migratory of the species, Smallmouth Buffalo, although this differentiation was found to be in adjacent river sections, rather than those separated by the greatest distance.

Genetic variation among Paddlefish and Smallmouth Buffalo populations in different river sections was not detected using AMOVA, and most genetic variation was attributed to within individuals. The F_{ST} statistics in general were very low for both species in this study, suggesting that there is sufficient gene flow to maintain similar allele frequencies and to avoid harmful effects of local inbreeding (Lowe and Allendorf 2010). Only one pairwise comparison was significant for Smallmouth Buffalo (between MFR and JBR), providing evidence that Smallmouth Buffalo may be experiencing some genetic differentiation between these regions; as such, these dams may be impacting Smallmouth Buffalo population genetics in this system slightly more than that of Paddlefish. Population structure analysis confirmed similar results to differentiation, in that one genetic cluster for each species was identified from the multiple river sections. Finally, DAPC analysis also showed a general pattern of low genetic differentiation, similar to that observed in the other analyses showing no clear separation or clusters among river sections for either

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species. A lack of genetic structure differentiation between isolated populations of fish has also been documented for Red Lionfish *Pterois volitans* collected in the northwestern Atlantic Ocean and the Gulf of Mexico (Pérez-Portela et al. 2018). Additionally, Raeymaekers et al. (2009) suggested that even if genetics did not reveal differentiation, it does not mean that processes leading to loss of diversity are not occurring as populations may not yet have reached equilibrium since being separated by barriers.

While it is clear that dams can affect genetic diversity in fish populations (Yamamoto et al. 2004; Heggenes and Røed 2006), it has been suggested that for most fish taxa, 50–100 years is sufficient time for barrier-induced divergence to occur. For example, Brown Trout *Salmo trutta* has a generation time of 3.5 years, and previous microsatellite analyses revealed substantial population divergence when comparing contemporary Brown Trout samples after 94 years of isolation due to dam construction (~27 generations; Heggenes and Røed 2006). The authors attribute the observed population differentiation to genetic drift because of low population densities among examined Brown Trout populations. The large values of N_e in both Paddlefish and Smallmouth Buffalo suggest that these populations are less prone to genetic drift, and therefore it is much easier to maintain genetic homogeneity across years of isolation. For long-lived species, such as sturgeons (family Acipenseridae) and Paddlefish, the current genetic structure is more likely to reflect historical processes and natural impediments to gene flow, especially when sampled individuals are adults (Lloyd et al. 2013; McDougall et al. 2017). Population differentiation in lock-and-dam systems may be caused by local adaptation through natural selection, genetic drift, gene flow, anthropogenic interference, or a combination of these factors. Extensive simulation and population genetic modeling is needed to place time points on genetic divergence. Finally, it is believed that the minimum effective population size necessary for viable isolated populations may be approximately 100 individuals to avoid inbreeding depression and approximately 1,000 to maintain adaptive potential into the long term (Frankham et al. 2014). As such, effective population sizes in these four river sections of the Alabama River should be monitored going forward.

Management implications

Our findings support that Paddlefish reproduction is occurring in each river section of the Alabama River, as well as in at least one of its upper tributaries. In addition, Paddlefish populations appear to be functionally isolated in JBR and MFR but less so between CL and LAR, where the crested spillway represents a partial barrier to mixing. Isolation of Paddlefish among river sections does not appear to have occurred over a long enough time to allow for genetic differences to develop in this long-lived species, or perhaps sufficient downstream drift of juveniles occurs to reduce this potential, which warrants future study. However, there remains a potential for genetic divergence to occur over a longer time frame, the implications of which are unclear. Similar patterns were seen for Smallmouth Buffalo, although variable results reduced our ability to detect significant differences. In addition, similar isolation and genetic divergence could be occurring in shorter-lived species, although on a shorter time scale. When considering the broader aquatic community, interruption of movement also has implications for the movement of mussel glochidia by fish hosts, potentially leading to broader impacts on the aquatic community. Similarly, altered habitats can become less suitable for the native fish community, often resulting in their decline and a reduction in the critical ecosystem functions they provide. Finally, some species, including Paddlefish, can be important economically, providing further rationale for the continued desire to maintain robust sustainable populations. Given the breadth of ecological, genetic, and economic concerns, longer-term assessment of the genetic consequences of the Alabama River dams should be quantified (for both shorter-lived and longer-lived species), and the facilitation of fish movement past these dams, particularly at Millers Ferry Lock and Dam and Robert F. Henry Lock and Dam, should be explored to strategically provide increases in upstream passage ability sequentially over time.

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CONFLICT OF INTEREST STATEMENT

There is no conflict of interest declared in this document.

DATA AVAILABILITY STATEMENT

Data are available upon request from the corresponding author.

ETHICS STATEMENT

Handling of fish was conducted in accordance with American Fisheries Society's *Guidelines for the Use of Fishes in Research* (Use of Fishes in Research Committee 2014), and under Auburn University's Institutional Animal Care and Use Committee Protocol 2016-2939.

ORCID

Garret J. Kratina https://orcid.org/0000-0003-2521-4796
Dennis R. DeVries https://orcid.org/0000-0002-3831-0755

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